Beam Structured Signal Detection for HF Skywave Massive MIMO Communications

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***Abstract*—In this paper, we investigate signal detection for HF skywave massive multiple-input multiple-output (MIMO) communications with orthogonal frequency division multiplexing (OFDM) modulation. We first introduce beam based channel model (BBCM) in the space domain and reveal the sparsity of the channel in the space-beam domain. Based on the BBCM in the space domain, we propose a beam structured detector (BSD) for each subcarrier. Specifically, we prove that the space domain detector design can be transformed to that of a beam domain detector without sacrificing optimality, and the asymptotically optimal space domain detector is beam structured with a low-dimensional beam domain detector, thus significantly reducing the design and implementation complexities. Furthermore, we provide a beam selection criterion to choose the beams that are used for the BSD. Simulation results demonstrate the low complexity and satisfactory performance of the proposed detector.**

I. Introduction

Future wireless network is demanded for seamless global service coverage including remote and rural areas [1]. High frequency (HF) communications, typically in the 3 to 30 MHz frequency band, can achieve over-the-horizon transmission by ionospheric reflection and provide worldwide coverage with relatively low deployment cost [2]. However, due to the limited spectrum resource and complicated ionospheric conditions, conventional point-to-point HF communications suffer from the low data rates.

As a key technology for the fifth-generation cellular wireless communications and its future evolution, massive multiple-input multiple-output (MIMO) can significantly enhance the spectrum and energy efficiencies by deploying a large number of antennas at the base station (BS) and serving a number of users in the same time-frequency resource simultaneously. Recently, massive MIMO in combination with orthogonal frequency division multiplexing (OFDM) has been introduced into HF communications in [3]. Moreover, the channel acquisition and downlink (DL) robust precoding for HF skywave massive MIMO has been investigated in [4] and [5], respectively. It is demonstrated in [3]–[5] that massive MIMO can provide significant performance gain for HF communications.

Besides channel estimation and precoding, the signal detection also plays an important role in massive MIMO communications and has been investigated in terrestrial cellular massive MIMO systems. Due to the increasing number of BS antennas and users, a maximum likelihood (ML) detector [6], even if can achieve the optimal performance, suffers from the unaffordable computational complexity [7], similarly to several other nonlinear detectors [8], [9]. On the other hand, linear detectors are proposed to reduce the computational burden, such as zero-forcing (ZF) detector and minimum mean-squared error (MMSE) detector [10]. Nevertheless, they involve large-dimensional matrix inversion and multiplication in massive MIMO systems, which leads to the prohibitive computational complexity as the dimension gets large. Therefore, several approximate matrix inversion based methods [11], [12] and iterative techniques that can avoid a direct matrix inversion [13], [14] are proposed. Note that the massive MIMO channel typically exhibits sparsity in the beam domain, which has been used to design low-complexity detectors recently [15]–[18].

In this paper, we investigate the signal detection for HF skywave massive MIMO communications. We present beam based channel model (BBCM) in the space domain and reveal the sparsity of the space-beam domain channel. Based on the BBCM, we propose a beam structured detector (BSD). Specifically, we prove that the space domain detector can be designed by optimizing the beam domain detector without sacrificing optimality and the BSD with a low-dimensional beam domain detector is asymptotically optimal for a sufficiently large number of BS antennas, where the used beams relate to those that each user terminal (UT) occupies, thus reducing the design and implementation complexities.

II. System Model

A. System Configuration

We consider a HF skywave massive MIMO-OFDM system, where the BS deploys a uniform linear array (ULA) with M antennas and serves U single-antenna UTs. Let N_{c} denote the number of subcarriers, where N_{v} subcarriers are used to

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transmit data with the index set $\mathcal{K} = \{k_0, k_1, \cdots, k_{N_\mathrm{v}-1}\}$. Moreover, we denote N_g, Δ_f, $T_\mathrm{s} = \frac{1}{N_\mathrm{c}\Delta_f}$, and $T_\mathrm{sym} = (N_\mathrm{c}+N_\mathrm{g})\,T_\mathrm{s}$ as the length of the cyclic prefix (CP), the subcarrier spacing, the sampling interval, and the duration of an OFDM symbol with CP, respectively.

In HF skywave communications, the carrier frequency, f_c, needs to be adjusted according to different ionospheric conditions. Hence, the inter-antenna spacing, d, is set according to the highest used carrier frequency f_o instead of f_c, that is, $d = \lambda_\mathrm{o}/2$, where $\lambda_\mathrm{o} = c/f_\mathrm{o}$ is the wavelength, and c is the speed of light. Note that this differs from the conventional massive MIMO system, where the inter-antenna spacing d is typically set according to the carrier frequency f_c.

B. Channel Model

During the uplink (UL) transmission, the received data at the k-th subcarrier on the n-th OFDM symbol at the m-th antenna is given by

$$y_{m,n,k} = \sum_{u=0}^{U-1} h^{\mathrm{SF}}_{u,m,n,k} x_{u,n,k} + z_{m,n,k}, \tag{1}$$

where $m \in \{0, 1, \cdots, M-1\}$, $k \in \mathcal{K}$, $x_{u,n,k}$ denotes the transmitted data of UT u on the n-th OFDM symbol at the k-th subcarrier, $z_{m,n,k}$ is the additive white Gaussian noise with distribution $\mathcal{CN}\left(0, \sigma_\mathrm{z}^2\right)$, and $h^{\mathrm{SF}}_{u,m,n,k}$ is the channel frequency response at the k-th subcarrier on the n-th OFDM symbol between UT u and the m-th antenna at the BS, which is

$$h^{\mathrm{SF}}_{u,m,n,k} = \sum_{p=0}^{P_u-1} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p}nT_\mathrm{sym}} e^{-\bar{\jmath}2\pi f_\mathrm{c} m\Delta_\tau\Omega_{u,p}} \times e^{-\bar{\jmath}2\pi k\Delta_f\tau_{u,p}} e^{-\bar{\jmath}2\pi k\Delta_f m\Delta_\tau\Omega_{u,p}}. \tag{2}$$

where P_u is the number of paths between the BS and UT u. In (2), $\gamma_{u,p}$ is the complex-valued gain of the p-th path of UT u and given by $\gamma_{u,p} = \beta_{u,p}e^{\bar{\jmath}\varphi_{u,p}}$, where $\beta_{u,p}$ and $\varphi_{u,p}$ are the gain and initial phase, respectively, and $\varphi_{u,p}$ is uniformly distributed over $[0, 2\pi)$. $\nu_{u,p}$ is the Doppler frequency, which is mainly caused by the UT mobility and the ionosphere. $\Omega_{u,p} \triangleq \sin\theta^{\mathrm{az}}_{u,p}\cos\theta^{\mathrm{el}}_{u,p}$ is the directional cosine, where $\theta^{\mathrm{az}}_{u,p}$ and $\theta^{\mathrm{el}}_{u,p}$ are the azimuth angle of arrival (AoA) and elevation AoA, respectively. $\tau_{u,p}$ is the p-th propagation path delay between UT u and the first antenna of the BS, and $\Delta_\tau = d/c$.

Let $\mathbf{h}^{\mathrm{S}}_{u,n,k} \in \mathbb{C}^{M\times 1}$ denote the space domain channel vector at the k-th subcarrier on the n-th OFDM symbol of UT u, with the m-th element, $h^{\mathrm{SF}}_{u,m,n,k}$. Therefore, according to (2), we have

$$\mathbf{h}^{\mathrm{S}}_{u,n,k} = \sum_{p=0}^{P_u-1} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p}nT_\mathrm{sym}} e^{-\bar{\jmath}2\pi k\Delta_f\tau_{u,p}} \mathbf{v}_k\left(\Omega_{u,p}\right), \tag{3}$$

where

$$\mathbf{v}_k(\Omega) \triangleq \left[1, e^{-\bar{\jmath}2\pi(f_\mathrm{c}+k\Delta_f)\Delta_\tau\Omega}, \cdots, e^{-\bar{\jmath}2\pi(f_\mathrm{c}+k\Delta_f)(M-1)\Delta_\tau\Omega}\right]^{\mathrm{T}}, \tag{4}$$

is the steering vector in the space domain, pointing towards directional cosine Ω. Note that due to the spatial-wideband effect caused by the increased bandwidth and equipped large-scale arrays [3], [4], [19], the steering vectors in the space domain differ for various subcarriers.

Note that the parameter $\Omega_{u,p}$ of each path for each UT is limited within set $\mathcal{B}_{\mathrm{an}} = \{\Omega\,|\Omega \in [-1, 1)\}$. We uniformly divide this set into multiple disjoint subsets as

$$\mathcal{B}_{\mathrm{an}} = \bigcup_{n_{\mathrm{an}}=0}^{N_{\mathrm{an}}-1} \Lambda^{\mathrm{an}}_{n_{\mathrm{an}}},\ \Lambda^{\mathrm{an}}_{n_{\mathrm{an}}} = \left[\frac{2n_{\mathrm{an}}-N_{\mathrm{an}}}{N_{\mathrm{an}}}, \frac{2n_{\mathrm{an}}-N_{\mathrm{an}}+2}{N_{\mathrm{an}}}\right), \tag{5}$$

The numbers of divided subsets for the directional cosine is $N_{\mathrm{an}} = \left\lceil F_{\mathrm{an}} M \frac{f_\mathrm{c}}{f_\mathrm{o}} \right\rceil$, where F_{an} is referred to as fine factor. Note that when N_{an} is sufficiently large, we can approximate the space steering vector $\mathbf{v}_k\left(\Omega_{u,p}\right)$ for $\Omega_{u,p} \in \Lambda^{\mathrm{an}}_{n_{\mathrm{an}}}$ as the sampled space steering vector $\mathbf{v}_k\left(\Omega_{n_{\mathrm{an}}}\right)$, where $\Omega_{n_{\mathrm{an}}} = \frac{2n_{\mathrm{an}}-N_{\mathrm{an}}}{N_{\mathrm{an}}}$. Therefore, denoting $\Gamma^{\mathrm{S}}_u \triangleq \{\Omega_{u,0}, \Omega_{u,1}, \cdots, \Omega_{u,P_u-1}\}$, the space domain channel vector, $\mathbf{h}^{\mathrm{S}}_{u,n,k}$, becomes

$$\mathbf{h}^{\mathrm{S}}_{u,n,k} = \sum_{n_{\mathrm{an}}=0}^{N_{\mathrm{an}}-1} \tilde{h}^{\mathrm{S}}_{u,n,k,n_{\mathrm{an}}} \mathbf{v}_k\left(\Omega_{n_{\mathrm{an}}}\right), \tag{6}$$

$$\tilde{h}^{\mathrm{S}}_{u,n,k,n_{\mathrm{an}}} = \sum_{\Omega_{n_{\mathrm{an}}} \in \Gamma^{\mathrm{S}}_u \cap \Lambda^{\mathrm{an}}_{n_{\mathrm{an}}}} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p}nT_\mathrm{sym}} e^{-\bar{\jmath}2\pi k\Delta_f\tau_{u,p}}. \tag{7}$$

Note that in (6), all the UTs share the same set of sampled space steering vectors. Each sampled space steering vector corresponds to a physical beam in the space domain. Therefore, we refer to (6) as the BBCM in the space domain, where $\tilde{h}^{\mathrm{S}}_{u,n,k,n_{\mathrm{an}}}$ is the beam domain channel coefficient.

Denote $\tilde{\mathbf{h}}^{\mathrm{S}}_{u,n,k}$ as the scaled space-beam domain channel with the n_{an}-th element, $\sqrt{M}\tilde{h}^{\mathrm{S}}_{u,n,k,n_{\mathrm{an}}}$, and define $\mathbf{V}_k = \frac{1}{\sqrt{M}}\left[\mathbf{v}_k\left(\Omega_0\right), \mathbf{v}_k\left(\Omega_1\right), \cdots, \mathbf{v}_k\left(\Omega_{N_{\mathrm{an}}-1}\right)\right] \in \mathbb{C}^{M\times N_{\mathrm{an}}}$. Then (6) can be

$$\mathbf{h}^{\mathrm{S}}_{u,n,k} = \mathbf{V}_k \tilde{\mathbf{h}}^{\mathrm{S}}_{u,n,k}, \tag{8}$$

where each column of $\mathbf{V}_k$ corresponds to a space-beam. In HF skywave massive MIMO communications, due to the small angle spread and limited propagation path number, the beam domain channels typically exhibit sparsity [4], which means that most elements of space-beam domain channels are approximately zero. The ordered index set of non-zero elements of the space-beam domain channel for UT u is defined as $\mathcal{N}^{\mathrm{an}}_u = \left\{n_{\mathrm{an}} \left| \left[\tilde{\mathbf{h}}^{\mathrm{S}}_{u,n,k}\right]_{n_{\mathrm{an}}} \neq 0 \right.\right\}$ with $\bar{N}_{\mathrm{an},u} = |\mathcal{N}^{\mathrm{an}}_u|$, where the order is the index increasing order. Define a ordered set $\mathcal{N}^{\mathrm{an}} = \bigcup_{u=0}^{U-1} \mathcal{N}^{\mathrm{an}}_u$ with $\bar{N}_{\mathrm{an}} = |\mathcal{N}^{\mathrm{an}}|$. Denoting $\bar{\mathbf{V}}_k = [\mathbf{V}_k]_{:,\mathcal{N}^{\mathrm{an}}}$, (8) can be rewritten as

$$\mathbf{h}^{\mathrm{S}}_{u,n,k} = \bar{\mathbf{V}}_k \bar{\mathbf{h}}^{\mathrm{S}}_{u,n,k}, \tag{9}$$

where $\bar{\mathbf{h}}^{\mathrm{S}}_{u,n,k} = \left[\tilde{\mathbf{h}}^{\mathrm{S}}_{u,n,k}\right]_{\mathcal{N}^{\mathrm{an}}}$, which shows that $\bar{N}_{\mathrm{an}}$ space-beams are sufficient to represent the space domain channel.

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III. Beam Structured Detector

A. Problem Formulation

We consider the UL transmission at the k-th subcarrier on the n-th OFDM symbol and omit index n in the subscript hereafter for brevity. The received signal in (1) can be rewritten in vector form as

$$\mathbf{y}_k = \mathbf{H}_k^{\mathrm{S}}\mathbf{x}_k + \mathbf{z}_k, \tag{10}$$

where $\mathbf{y}_k = [y_{0,k}, y_{1,k}, \cdots, y_{M-1,k}]^{\mathrm{T}} \in \mathbb{C}^{M\times 1}$, $\mathbf{H}_k^{\mathrm{S}} = \left[\mathbf{h}_{0,k}^{\mathrm{S}}, \mathbf{h}_{1,k}^{\mathrm{S}}, \cdots, \mathbf{h}_{U-1,k}^{\mathrm{S}}\right]$, $\mathbf{x}_k = [x_{0,k}, x_{1,k}, \cdots, x_{U-1,k}]^{\mathrm{T}} \in \mathbb{C}^{U\times 1}$, and $\mathbf{z}_k$ is the additive white Gaussian noise with distribution $\mathcal{CN}\left(\mathbf{0}, \sigma_{\mathrm{z}}^2\mathbf{I}_M\right)$. We assume that $\mathbb{E}\{\mathbf{x}_k\mathbf{x}_k^{\mathrm{H}}\} = \sigma_{\mathrm{x}}^2\mathbf{I}_U$, where σ_{x}^2 is the average transmit power of each UT.

Consider a linear detector $\mathbf{w}_{u,k} \in \mathbb{C}^{M\times 1}$ for UT u and define $\mathbf{W}_k = [\mathbf{w}_{0,k}, \mathbf{w}_{1,k}, \cdots, \mathbf{w}_{U-1,k}] \in \mathbb{C}^{M\times U}$. Then the MSE of the detection result is given by

$$\mathrm{MSE}_k = \mathbb{E}\left\{\left\|\mathbf{x}_k - \mathbf{W}_k^{\mathrm{H}}\mathbf{y}_k\right\|^2\right\}. \tag{11}$$

Thus the conventional space domain MMSE detector is

$$\mathbf{W}_k^{\mathrm{mmse}} = \arg\min_{\mathbf{W}_k} \ \mathrm{MSE}_k = \mathbf{H}_k^{\mathrm{S}}\left(\left(\mathbf{H}_k^{\mathrm{S}}\right)^{\mathrm{H}}\mathbf{H}_k^{\mathrm{S}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}_U\right)^{-1}, \tag{12}$$

and the detection result is

$$\hat{\mathbf{x}}_k = (\mathbf{W}_k^{\mathrm{mmse}})^{\mathrm{H}}\mathbf{y}_k. \tag{13}$$

In HF skywave massive MIMO-OFDM communications, with increase number of the BS antennas and UTs, the large M and U will make the computation of $\mathbf{W}_k^{\mathrm{mmse}}$ very inefficient in practical systems due to the large-dimensional matrix inversion and multiplication. However, the angle spread of the HF skywave channel is typically small [4], which means that the space-beam domain channel has the significant sparsity and the channel energy of each UT is concentrated on a few non-zero beams. Therefore, this motivates us to utilize such a sparsity in beam domain to design a structured detector.

B. Beam Structured Detector

We consider the UL detector design for UT u, whose MSE for signal detection is

$$\begin{aligned}\mathrm{MSE}_{u,k} &= \mathbb{E}\left\{\left\|x_{u,k} - \mathbf{w}_{u,k}^{\mathrm{H}}\mathbf{y}_k\right\|^2\right\}\\ &= \sigma_{\mathrm{x}}^2 - \sigma_{\mathrm{x}}^2\mathbf{w}_{u,k}^{\mathrm{H}}\mathbf{h}_{u,k}^{\mathrm{S}} - \sigma_{\mathrm{x}}^2\left(\mathbf{h}_{u,k}^{\mathrm{S}}\right)^{\mathrm{H}}\mathbf{w}_{u,k}\\ &\quad + \sigma_{\mathrm{x}}^2\sum_{u'=0}^{U-1}\mathbf{w}_{u,k}^{\mathrm{H}}\mathbf{h}_{u',k}^{\mathrm{S}}\left(\mathbf{h}_{u',k}^{\mathrm{S}}\right)^{\mathrm{H}}\mathbf{w}_{u,k} + \sigma_{\mathrm{z}}^2\mathbf{w}_{u,k}^{\mathrm{H}}\mathbf{w}_{u,k}.\end{aligned} \tag{14}$$

Then the design of detector $\mathbf{w}_{u,k}$ for UT u can be formulated as the following problem

$$\mathbf{w}_{u,k}^{\mathrm{op}} = \arg\min_{\mathbf{w}_{u,k}} \mathrm{MSE}_{u,k}. \tag{15}$$

Motivated by sparse BBCM (9), we rewrite $\mathrm{MSE}_{u,k}$ as

$$\begin{aligned}\mathrm{MSE}_{u,k} &= \sigma_{\mathrm{x}}^2 - \sigma_{\mathrm{x}}^2\tilde{\mathbf{w}}_{u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u,k}^{\mathrm{S}} - \sigma_{\mathrm{x}}^2\left(\bar{\mathbf{h}}_{u,k}^{\mathrm{S}}\right)^{\mathrm{H}}\tilde{\mathbf{w}}_{u,k}\\ &\quad + \sigma_{\mathrm{x}}^2\sum_{u'=0}^{U-1}\tilde{\mathbf{w}}_{u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\left(\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\right)^{\mathrm{H}}\tilde{\mathbf{w}}_{u,k} + \sigma_{\mathrm{z}}^2\mathbf{w}_{u,k}^{\mathrm{H}}\mathbf{w}_{u,k},\end{aligned} \tag{16}$$

where $\tilde{\mathbf{w}}_{u,k} = \bar{\mathbf{V}}_k^{\mathrm{H}}\mathbf{w}_{u,k} \in \mathbb{C}^{\bar{N}_{\mathrm{an}}\times 1}$. In (16), the space-beam domain channel $\bar{\mathbf{h}}_{u,k}^{\mathrm{S}}$ instead of the space domain channel $\mathbf{h}_{u,k}^{\mathrm{S}}$ is involved and $\mathbf{w}_{u,k}$ in most terms is replaced by $\tilde{\mathbf{w}}_{u,k}$. Although $\mathbf{w}_{u,k}$ is still involved in the term $\sigma_{\mathrm{z}}^2\mathbf{w}_{u,k}^{\mathrm{H}}\mathbf{w}_{u,k}$, it can be obtained according to the relation $\tilde{\mathbf{w}}_{u,k} = \bar{\mathbf{V}}_k^{\mathrm{H}}\mathbf{w}_{u,k}$ when $\tilde{\mathbf{w}}_{u,k}$ is given. Note that for any given $\tilde{\mathbf{w}}_{u,k}$, the sufficient and necessary condition for the existence of $\mathbf{w}_{u,k}$ such that $\tilde{\mathbf{w}}_{u,k} = \bar{\mathbf{V}}_k^{\mathrm{H}}\mathbf{w}_{u,k}$ is $\left(\bar{\mathbf{V}}_k^H\left(\bar{\mathbf{V}}_k^H\right)^{\dagger} - \mathbf{I}\right)\tilde{\mathbf{w}}_{u,k} = \mathbf{0}$, where the superscript $\dagger$ denotes the Moore-Penrose inverse. Hence, we can turn to optimize $\tilde{\mathbf{w}}_{u,k}$ instead of $\mathbf{w}_{u,k}$ to minimize the MSE. We can obtain the following theorem, whose proof is omitted due to space limitations.

Theorem 1: The optimal solution of problem in (15) can be obtained as

$$\mathbf{w}_{u,k}^{\mathrm{op}} = \bar{\mathbf{V}}_k\left(\bar{\mathbf{V}}_k^{\mathrm{H}}\bar{\mathbf{V}}_k\right)^{\dagger}\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}, \tag{17}$$

where

$$\begin{aligned}\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}} = \underset{\tilde{\mathbf{w}}_{u,k}}{\operatorname{argmin}}\ & \sigma_{\mathrm{x}}^2 - \sigma_{\mathrm{x}}^2\tilde{\mathbf{w}}_{u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u,k}^{\mathrm{S}} - \sigma_{\mathrm{x}}^2\left(\bar{\mathbf{h}}_{u,k}^{\mathrm{S}}\right)^{\mathrm{H}}\tilde{\mathbf{w}}_{u,k}\\ &+ \tilde{\mathbf{w}}_{u,k}^{\mathrm{H}}\left(\sigma_{\mathrm{x}}^2\sum_{u'=0}^{U-1}\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\left(\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\right)^{\mathrm{H}} + \sigma_{\mathrm{z}}^2\left(\bar{\mathbf{V}}_k^{\mathrm{H}}\bar{\mathbf{V}}_k\right)^{\dagger}\right)\tilde{\mathbf{w}}_{u,k}\\ \text{s.t.}\ & \left(\bar{\mathbf{V}}_k^{\mathrm{H}}\left(\bar{\mathbf{V}}_k^{\mathrm{H}}\right)^{\dagger} - \mathbf{I}\right)\tilde{\mathbf{w}}_{u,k} = \mathbf{0}.\end{aligned} \tag{18}$$

Theorem 1 provides an optimal detector structure. When we obtain $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}$ by solving the optimization problem in (18), we can directly construct detector $\mathbf{w}_{u,k}^{\mathrm{op}}$ according to the relation (17). Moreover, we assume that the number of UTs is limited and the directional cosines of the paths of UTs are distinct [3]. Thus when $M \to \infty$, the columns of $\bar{\mathbf{V}}_k$ are asymptotically orthogonal and it can be checked that $\lim_{M\to\infty}\bar{\mathbf{V}}_k^H\left(\bar{\mathbf{V}}_k^H\right)^{\dagger} - \mathbf{I} = \mathbf{0}$ and $\lim_{M\to\infty}\left(\bar{\mathbf{V}}_k^{\mathrm{H}}\bar{\mathbf{V}}_k\right)^{\dagger} = \mathbf{I}$. In this case, the optimal solution in (17) becomes

$$\mathbf{w}_{u,k}^{\mathrm{op}} = \bar{\mathbf{V}}_k\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}, \tag{19}$$

where

$$\begin{aligned}\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}} = \underset{\tilde{\mathbf{w}}_{u,k}}{\operatorname{argmin}}\ & \sigma_{\mathrm{x}}^2 - \sigma_{\mathrm{x}}^2\tilde{\mathbf{w}}_{u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u,k}^{\mathrm{S}} - \sigma_{\mathrm{x}}^2\left(\bar{\mathbf{h}}_{u,k}^{\mathrm{S}}\right)^{\mathrm{H}}\tilde{\mathbf{w}}_{u,k}\\ &+ \tilde{\mathbf{w}}_{u,k}^{\mathrm{H}}\left(\sigma_{\mathrm{x}}^2\sum_{u'=0}^{U-1}\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\left(\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\right)^{\mathrm{H}} + \sigma_{\mathrm{z}}^2\mathbf{I}\right)\tilde{\mathbf{w}}_{u,k}.\end{aligned} \tag{20}$$

Relation in (19) provides an asymptotically optimal detector structure, which is much simpler than (17). Meanwhile, the optimization problem in (20) also has a simpler form. Since only the space-beam domain channels are involved in (20) and the structure in (19) is very similar to that of BBCM (9), we refer to $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}$ as the beam domain detector. The resulting space domain detector, $\mathbf{w}_{u,k}^{\mathrm{op}}$, is referred to as the BSD.

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In the optimization problem in (20), the dimension of optimization variable $\tilde{\mathbf{w}}_{u,k}$ is still large since it considers non-zero beams of all UTs and the search for $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}$ is still computationally expensive. Thus we have the following theorem, which can significantly reduce the dimension of $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}$. The proof is omitted due to space limitations.

Theorem 2: When $\forall u, u', \mathcal{N}_u^{\mathrm{an}} \cap \mathcal{N}_{u'}^{\mathrm{an}} = \emptyset$, the optimal solution of (20) satisfies

$$\left[\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}\right]_{n_{\mathrm{an}}} = 0, \text{ for } [\mathcal{N}^{\mathrm{an}}]_{n_{\mathrm{an}}} \notin \mathcal{N}_u^{\mathrm{an}}. \tag{21}$$

From Theorem 2, when there are no overlapping directional cosine sets for any two UTs, the elements of $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}$ are non-zero only in the positions corresponding to the non-zero beams that UT u occupies, for a sufficiently large M. Due to the sparsity of the space-beam domain channel for HF skywave massive MIMO communications, the number of non-zero beams of UT u, $\bar{N}_{\mathrm{an},u}$, is much smaller than the number of the total beams N_{an} and antennas M. Thus we focus on the non-zero elements of $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}$ and the dimensions of the optimization variable can be much smaller. Therefore, the optimal solution in (19) can be rewritten as

$$\mathbf{w}_{u,k}^{\mathrm{op}} = \mathbf{V}_k \mathbf{S}_u \bar{\mathbf{w}}_{u,k}^{\mathrm{op}}, \tag{22}$$

where

$$\begin{aligned}\bar{\mathbf{w}}_{u,k}^{\mathrm{op}} = \underset{\bar{\mathbf{w}}_{u,k}}{\operatorname{argmin}}\, & \sigma_{\mathrm{x}}^2 - \sigma_{\mathrm{x}}^2 \bar{\mathbf{w}}_{u,k}^{\mathrm{H}} \mathbf{S}_u^{\mathrm{H}} \tilde{\mathbf{h}}_{u,k}^{\mathrm{S}} - \sigma_{\mathrm{x}}^2 \left(\tilde{\mathbf{h}}_{u,k}^{\mathrm{S}}\right)^{\mathrm{H}} \mathbf{S}_u \bar{\mathbf{w}}_{u,k} \\ & + \bar{\mathbf{w}}_{u,k}^{\mathrm{H}} \left(\sigma_{\mathrm{x}}^2 \sum_{u'=0}^{U-1} \mathbf{S}_u^{\mathrm{H}} \tilde{\mathbf{h}}_{u',k}^{\mathrm{S}} \left(\tilde{\mathbf{h}}_{u',k}^{\mathrm{S}}\right)^{\mathrm{H}} \mathbf{S}_u + \sigma_{\mathrm{z}}^2 \mathbf{I}\right) \bar{\mathbf{w}}_{u,k} \\ = & \left(\sum_{u'=0}^{U-1} \mathbf{S}_u^{\mathrm{H}} \tilde{\mathbf{h}}_{u',k}^{\mathrm{S}} \left(\tilde{\mathbf{h}}_{u',k}^{\mathrm{S}}\right)^{\mathrm{H}} \mathbf{S}_u + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2} \mathbf{I}\right)^{-1} \mathbf{S}_u^{\mathrm{H}} \tilde{\mathbf{h}}_{u,k}^{\mathrm{S}}, \end{aligned} \tag{23}$$

and $\mathbf{S}_u \in \mathbb{R}^{N_{\mathrm{an}} \times \bar{N}_{\mathrm{an},u}}$ is the beam selection martrix with

$$[\mathbf{S}_u]_{i,j} = \begin{cases} 1, & i = [\mathcal{N}_u^{\mathrm{an}}]_j \\ 0, & \text{otherwise.} \end{cases} \tag{24}$$

and $\bar{\mathbf{w}}_{u,k}^{\mathrm{op}}$ is the low-dimensional beam domain detector corresponding to the non-zeros elements of $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}$. For brevity, we denote the BSD in (22) with the beam index set $\mathcal{N}_u^{\mathrm{an}}$ as $\mathbf{w}_{u,k}^{\mathrm{op}}(\mathcal{N}_u^{\mathrm{an}})$.

When the number of UTs becomes larger, the condition that the directional cosine sets for any two UTs are non-overlapping, however, may not be well satisfied. In this case, more beams need to be considered to improve the detection performance. For this purpose, we have the following Theorem, where $\mathrm{MSE}_{u,k}\left(\mathbf{w}_{u,k}^{\mathrm{op}}\left(\mathcal{N}_{u,1}^{\mathrm{an}}\right)\right)$ and $\mathrm{MSE}_{u,k}\left(\mathbf{w}_{u,k}^{\mathrm{op}}\left(\mathcal{N}_{u,2}^{\mathrm{an}}\right)\right)$ refer to the MSEs of detectors, $\mathbf{w}_{u,k}^{\mathrm{op}}\left(\mathcal{N}_{u,1}^{\mathrm{an}}\right)$ and $\mathbf{w}_{u,k}^{\mathrm{op}}\left(\mathcal{N}_{u,2}^{\mathrm{an}}\right)$, respectively. The proof is omitted due to space limitations.

Theorem 3: For a given beam index set $\mathcal{N}_{u,1}^{\mathrm{an}} \supseteq \mathcal{N}_u^{\mathrm{an}}$, define a new beam index set $\mathcal{N}_{u,2}^{\mathrm{an}} = \mathcal{N}_{u,1}^{\mathrm{an}} \cup \{\tilde{n}_{\mathrm{an}}\}$ where $\tilde{n}_{\mathrm{an}} \notin \mathcal{N}_{u,1}^{\mathrm{an}}$. When $M \to \infty$, we have

$$\mathrm{MSE}_{u,k}\left(\mathbf{w}_{u,k}^{\mathrm{op}}\left(\mathcal{N}_{u,1}^{\mathrm{an}}\right)\right) \geq \mathrm{MSE}_{u,k}\left(\mathbf{w}_{u,k}^{\mathrm{op}}\left(\mathcal{N}_{u,2}^{\mathrm{an}}\right)\right), \tag{25}$$

where equality holds provided that

$$\left[\mathbf{H}_k^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:} \left[\mathbf{H}_k^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}} = \mathbf{0}. \tag{26}$$

From Theorem 3, when the directional set of UT u overlaps with those of other UTs, it is not optimal for the BSD design with only beams occupied by UT u and more beams need to be used for the detector design. Moreover, Theorem 3 provides a criterion for the selection of the additional beams. The beams that satisfy the condition (26) should not be selected since it will not improve the performance of the detector, while the beams that satisfy the condition $[\mathbf{H}_k^{\mathrm{S}}]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}[\mathbf{H}_k^{\mathrm{S}}]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}} \neq \mathbf{0}$ should be selected to improve the performance of the BSD. Here, we illustrate it in Fig 1 as an example. As shown in Fig 1, when the beam set of UT 2 overlaps with that of UTs 1 and 3, we can select another two beams in addition to the beams occupied by UT 2 as shown by the solid line box to improve the performance of the detector. Moreover, we can add another two beams as shown in dotted line box to further reduce the MSE and improve the performance. According to Theorem 3, if we want to achieve the optimal detector performance, all the beams of the UT whose beam index set overlaps with the already selected beam index set should be selected. However, as shown in the simulation section, satisfactory performance can be achieved by selecting only several beams around the UT in addition to the beams it occupies.

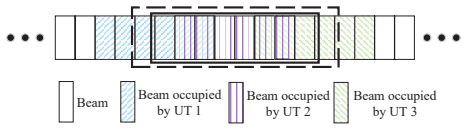


Fig. 1. Illustration of Theorem 3.

Although the number of antennas in practical systems is finite, the above discussion suggests that the BSD still can be used to achieve near-optimal performance and low complexity when the number of BS antennas is large. Therefore, we constrain the space domain detector of UT u to have the structure $\mathbf{w}_{u,k} = \mathbf{V}_k \hat{\mathbf{S}}_u \hat{\mathbf{w}}_{u,k}$, where the definition of the beam selection matrix $\hat{\mathbf{S}}_u$ is similar to $\mathbf{S}_u$ except that the ordered selected beam index set, $\mathcal{N}_u^{\mathrm{an}}$, is replaced with $\hat{\mathcal{N}}_u^{\mathrm{an}} \supseteq \mathcal{N}_u^{\mathrm{an}}$ and $\left|\hat{\mathcal{N}}_u^{\mathrm{an}}\right| = \hat{N}_{\mathrm{an},u}$. Thus we turn to optimize $\hat{\mathbf{w}}_{u,k}$, whose optimal solution in terms of MSE minimization is

$$\begin{aligned}\hat{\mathbf{w}}_{u,k}^{\mathrm{B}} &= \underset{\hat{\mathbf{w}}_{u,k}}{\arg\min}\, \mathbb{E}\left\{\left\| x_{u,k} - \hat{\mathbf{w}}_{u,k}^{\mathrm{H}} \hat{\mathbf{S}}_u^{\mathrm{H}} \mathbf{V}_k^{\mathrm{H}} \mathbf{y}_k \right\|^2\right\} \\ &= \left(\sum_{u'=0}^{U-1} \mathbf{P}_{u,u',k} \hat{\mathbf{h}}_{u',k}^{\mathrm{S}} \left(\hat{\mathbf{h}}_{u',k}^{\mathrm{S}}\right)^{\mathrm{H}} \mathbf{P}_{u',u,k} + \mathbf{P}_{u,u,k}\right)^{-1} \mathbf{P}_{u,u,k} \hat{\mathbf{h}}_{u,k}^{\mathrm{S}}, \end{aligned} \tag{27}$$

where $\mathbf{P}_{u,u',k} \triangleq \hat{\mathbf{S}}_u^{\mathrm{H}} \mathbf{V}_k^{\mathrm{H}} \mathbf{V}_k \hat{\mathbf{S}}_{u'} \in \mathbb{C}^{\hat{N}_{\mathrm{an},u} \times \hat{N}_{\mathrm{an},u'}}$ and $\hat{\mathbf{h}}_{u,k}^{\mathrm{S}} \in \mathbb{C}^{\hat{N}_{\mathrm{an},u} \times 1}$ is the low-dimensional space-beam domain channel corresponding to the beam index set $\hat{\mathcal{N}}_u^{\mathrm{an}}$. Then the space domain detector is

$$\mathbf{w}_{u,k}^{\mathrm{B}} = \mathbf{V}_k \hat{\mathbf{S}}_u \hat{\mathbf{w}}_{u,k}^{\mathrm{B}}, \tag{28}$$

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and the detection result is

$$\hat{x}_u = \left(\mathbf{w}_{u,k}^{\mathrm{B}}\right)^{\mathrm{H}}\mathbf{y}=\left(\hat{\mathbf{w}}_{u,k}^{\mathrm{B}}\right)^{\mathrm{H}}\hat{\mathbf{S}}_u^{\mathrm{H}}\mathbf{V}_k^{\mathrm{H}}\mathbf{y}_k. \tag{29}$$

Note that the beam index set, $\hat{\mathcal{N}}_u^{\mathrm{an}}$, can be flexibly selected to trade off the performance and complexity. Moreover, when M tends to infinity and $\hat{\mathcal{N}}_u^{\mathrm{an}}=\mathcal{N}_u^{\mathrm{an}}$, (27) will become (23).

C. Complexity Analysis

We analyze the complexity of the proposed BSD in this subsection. For simplicity, we define the average value and the maximum value of $\hat{N}_{\mathrm{an},u}$ as $\hat{N}_{\mathrm{an}}^{\mathrm{ave}} = \frac{1}{U}\sum_{u=0}^{U-1}\hat{N}_{\mathrm{an},u}$ and $\hat{N}_{\mathrm{an}}^{\max}=\max\left(\hat{N}_{\mathrm{an},u}\right)$, respectively. We first analyze the design complexity of the detector, which comes from (27). We can find that only low-dimensional space-beam domain channels $\hat{\mathbf{h}}_{0,k}^{\mathrm{S}},\cdots,\hat{\mathbf{h}}_{U-1,k}^{\mathrm{S}}$ are involved in (27), and the dimension of matrix inversion is greatly reduced and depends on the number of selected beams. Moreover, the matrix $\mathbf{P}_{u,u',k}$ can be pre-calculated since it is related to the non-zero beam indexes, which rely on the statistical CSI and vary slowly [4], [19], [20]. Therefore, the design complexity of the proposed detector in terms of complex multiplication number is $\mathcal{O}\left(\left(\hat{N}_{\mathrm{an}}^{\max}\right)^2U^2+\left(\hat{N}_{\mathrm{an}}^{\max}\right)^3U\right)$, which is much less than the design complexity $\mathcal{O}\left(MU^2+U^3\right)$ of the space domain MMSE detector in (12) due to the significant sparsity in the beam domain of HF skywave massive MIMO-OFDM channels.

We then discuss the detection complexity with the proposed detector, which comes from (29). Note that $\mathbf{V}_k$ can be treated as a chirp z-transform (CZT) matrix and rewritten as [21]

$$\begin{aligned}\mathbf{V}_k =& \operatorname{diag}\left\{\boldsymbol{\xi}_{W_{\mathrm{S},k},N_{\mathrm{an}},0,M}\right\}\mathbf{F}_{N_{(\mathrm{S})}\times M}^{\mathrm{H}}\operatorname{diag}\left\{\tilde{\boldsymbol{\xi}}_{W_{\mathrm{S},k},M,N_{\mathrm{an}},N_{(\mathrm{S})}}\right\}\\ &\times\mathbf{F}_{N_{(\mathrm{S})}\times N_{\mathrm{an}}}\operatorname{diag}\left\{\boldsymbol{\xi}_{W_{\mathrm{S},k},0,0,N_{\mathrm{an}}}\right\},\end{aligned} \tag{30}$$

where $\mathbf{F}_N$ denotes the N-dimensional unitary discret Fourier transform (DFT) matrix, $\mathbf{F}_{N\times G}$ denotes the matrix composed of the first G ($\leq N$) columns of $\mathbf{F}_N$, $N_{(\mathrm{S})}$ is the integer greater than or equal to $M+N_{\mathrm{an}}-1$. $W_{\mathrm{S},k} = e^{-\bar{\jmath}4\pi(f_{\mathrm{c}}+k\Delta_f)\Delta_\tau/N_{\mathrm{an}}}$, $\boldsymbol{\xi}_{W,\alpha_1,\alpha_2,N}\in\mathbb{C}^{N\times 1}$ with $[\boldsymbol{\xi}_{W,\alpha_1,\alpha_2,N}]_i = W^{\left(i^2-\alpha_1 i-\alpha_2\right)/2}$, and $\tilde{\boldsymbol{\xi}}_{W,N_1,N_2,N}\triangleq\sqrt{N}\mathbf{F}_N\left[\boldsymbol{\xi}_{W,0,0,N_1}^{\mathrm{H}},\mathbf{0},\boldsymbol{\xi}_{W,2(N_2-1),-(N_2-1)^2,N_2-1}^{\mathrm{H}}\right]^{\mathrm{T}}\in\mathbb{C}^{N\times 1}$. Then the multiplication $\mathbf{V}_k^{\mathrm{H}}\mathbf{y}$ can be efficiently implemented and needs to be calculated only once for all UTs since it is not related with the specific UT. Thus the detection complexity of the proposed detector in terms of complex multiplication number is $\mathcal{O}\left(N_{(\mathrm{S})}\log_2 N_{(\mathrm{S})}+\hat{N}_{\mathrm{an}}^{\mathrm{ave}}U\right)$, while the detection complexity of the space domain MMSE detector is $\mathcal{O}(MU)$ according to (13).

IV. Simulation Results

In this section, we provide simulation results to illustrate the efficiency of the proposed detector for HF skywave massive MIMO-OFDM communications.

The main simulation parameters are given in Table I. We generate UTs at a distance of 2,000 km from the BS, and azimuth angles of the UTs seen from the BS are generated in the interval (-90° 90°). The realistic HF skywave channel parameters are generated for each UT independently by a commercial ray-tracing software Proplab-Pro version 3.1 [22]. Specifically, the software is utilized to generate parameters of each propagation path between the BS and each UT, including the path gain, the propagation delay, the azimuth and elevation AoA. The initial phases and Doppler frequencies for paths are generated randomly. Then using the generated path parameters for paths, the beam based channel in the space domain can be generated according to (6), where fine factor F_{an} is set to two [4]. Moreover, using a similar approach in [4], we schedule the UTs to select those with relatively low channel correlation for simulation. The signal-to-noise ratio (SNR) in the simulation refers to the received SNR. The modulation scheme is 16 quadrature amplitude modulation (QAM) and the hard decision is used.

TABLE I
Simulation Parameters

Parameter	Value	Parameter	value
Carrier frequency f_{c}	16 MHz	BS antenna spacing d	9 m
Subcarrier number N_{c}	2048	Valid subcarrier number N_{v}	1536
Subcarrier spacing Δ_f	250 Hz	Modulation scheme	16-QAM

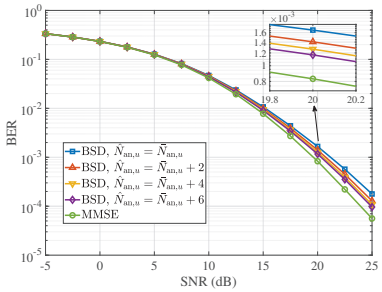


Fig. 2. Performance of BSD with different number of selected beams and MMSE detector.

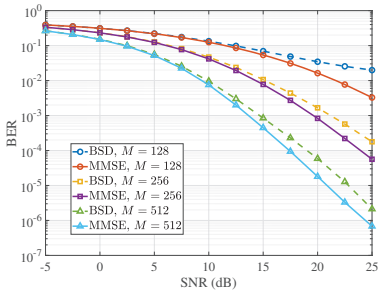


Fig. 3. Performance of BSD and MMSE detector with different number of BS antennas.

Fig. 2 shows the BER performance of the BSD with different number of selected beams, where $M=256$ and $U=96$. We select beams according to Theorem 3. From the figure, the performance of the BSD approaches that of the MMSE detector and the performance gap increases with SNR due to the inter-UT interference. However, the performance

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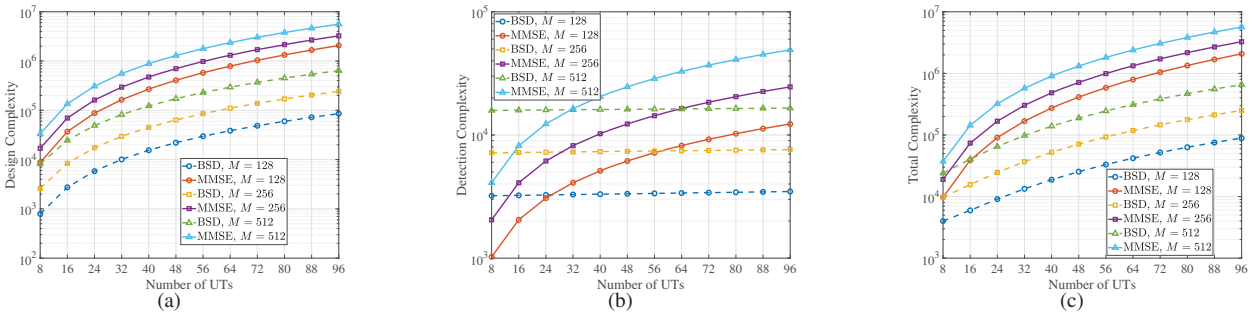


Fig. 4. The complexities of BSD and MMSE detector versus the number of UTs. (a) Design complexity; (b) Detection complexity; (c) Total complexity.

gap between the BSD and the MMSE detector decreases as the number of selected beams increases. This is because the performance of the BSD is further improved by using more beams in addition to those occupied by the UT itself to tackle the inter-UT interference according to Theorem 3.

Fig. 3 shows the BER performance of the BSD and the MMSE detector with different number of BS antennas, where $U = 96$ and $\hat{N}_{\mathrm{an},u} = \bar{N}_{\mathrm{an},u}$. From the figure, the performance gap between the BSD and the MMSE detector decreases as the number of BS antennas increases. This is because when M become large, different beams become approximately orthogonal, which reduces inter-UT interference and the performance gap. It can be expected that as M becomes sufficiently large, the performance gap will tend to disappear if the UTs are well scheduled to satisfy the condition that the directional cosine sets are non-overlapping according to Theorem 2.

Fig. 4 shows the complexity of different detectors, where $\hat{N}_{\mathrm{an},u} = \bar{N}_{\mathrm{an},u}$. The total complexity refers to the sum of the design complexity and the detection complexity. From the figures, the BSD has much lower complexity than MMSE due to the utilization of the sparsity of the beam domain channel, especially when the number of UTs becomes larger.

V. Conclusions

In this paper, we have investigated signal detection for HF skywave massive MIMO communications. We first introduced the BBCM and revealed the sparsity of the space-beam domain channel. Based on the BBCM, we proposed a BSD, where we proved that the space domain detector can be designed by optimizing a low-dimensional beam domain detector, and corresponding beam selection criterion was also provided. Furthermore, the proposed detector can be efficiently implemented by exploiting the beam structure. Simulation results verified the effectiveness of our proposed detector.

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